Orbit SIS

SSN College of Engineering, Chennai

System Design Poster

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Course System Design Lab

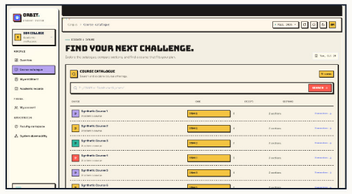
Orbit SIS: Student Information System for Enrollment and Academic Records

Introduction

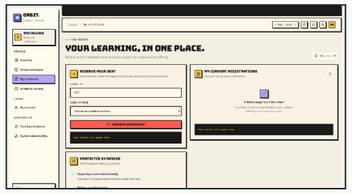
Orbit SIS is a full-stack student information system for course discovery, enrollment and academic records, with separate student and faculty workflows.

When several students request the last seat at nearly the same time, a check-then-write design lets two requests see the same seat. Here PostgreSQL is the single authority for seat capacity, and the decision is tested under concurrent load.

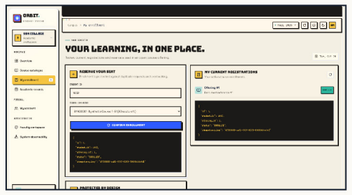
Interface



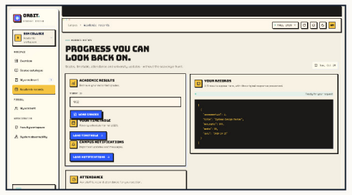
Course discovery



Enrollment form



Confirmed registration

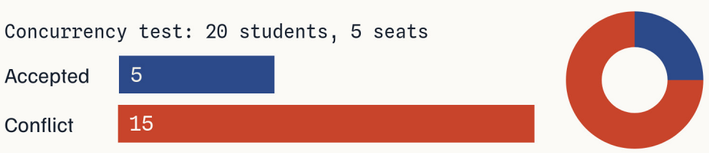


Academic records

Captured from the local application. All records are synthetic.

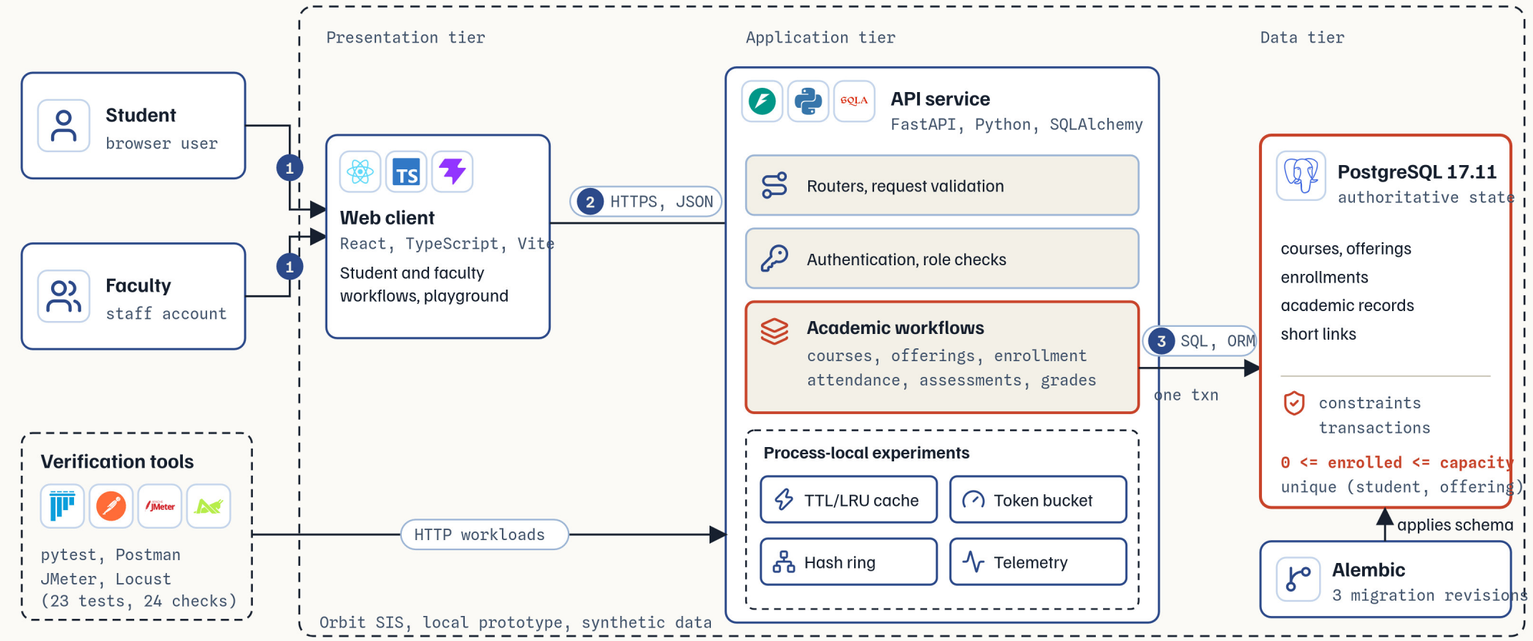
Results

Concurrency test: 20 students, 5 seats



PostgreSQL test suite	23 passed
Postman / Newman	24/24 assertions
JMeter, paced	185 samples, 0 errors
JMeter, unpaced	83.32% failed (HTTP 429)

Architecture Diagram



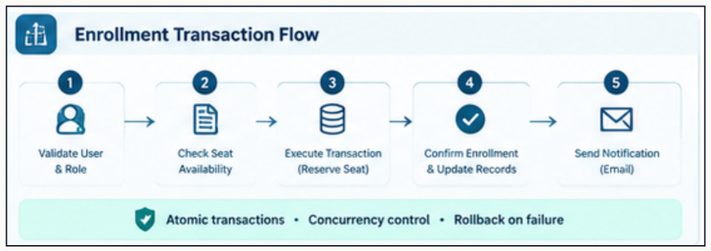
Container

Process-local or external

Enforces the correctness rule

Arrow: caller to callee

Data Flow & Technology



Seat check and reservation is one conditional UPDATE, then the INSERT in the same transaction. A full offering or a duplicate rolls it back (409).



Learning & Innovation

What we learned

- Checking capacity in application code leaves a race window, so the database has to make the decision.
- A uniqueness constraint protects stored data; an idempotency key protects client retries.
- JavaScript cannot hold every 64-bit ID as a number, so the API returns IDs as strings.
- Each test result is reported with its scope; a passing run proves only what it asserts.

What we built

- A conditional seat update and enrollment insert in one transaction.
- A TTL/LRU cache, token-bucket limiter, consistent hash ring and telemetry, each inspectable from a System Playground screen.